Queue operations using linked list

Module II

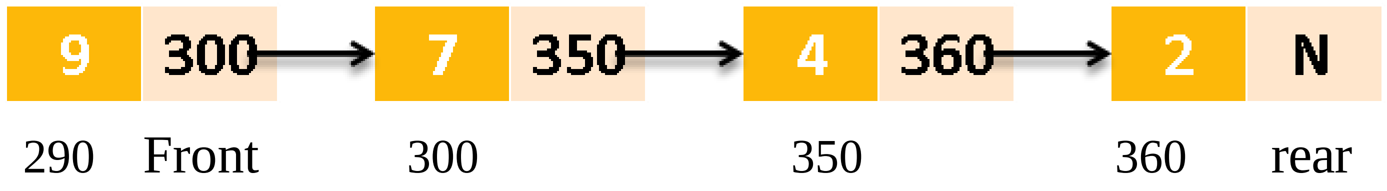
Dr. S. M. Farooq
Assistant Professor Senior
School of Computer Science and Engineering
Vellore Institute of Technology, Vellore

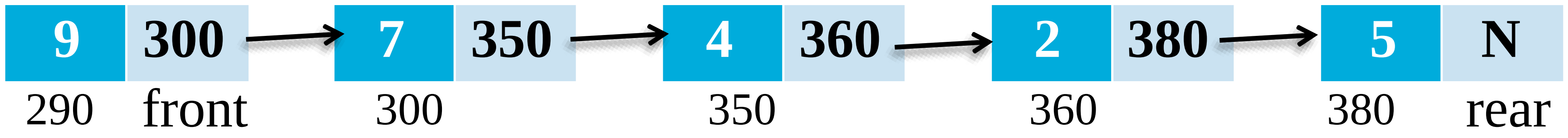
Representation of Queues

- Queues can be represented using arrays
- Queues can be represented using Linked List

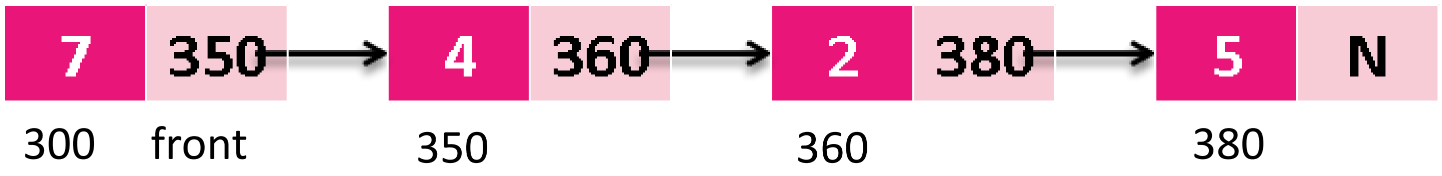
Array representation of Queues

Linked representation of Queues





Linked queue after inserting a node



Linked queue after deleting a node

FRONT=REAR=NULL

Insert an element in queue using linked list

STEP-1: Allocate memory for the new node & name it as TEMP.

STEP-2: SET TEMP→ DATA = NUM

SET TEMP→ LINK = NULL

STEP-3: IF FRONT = NULL

FRONT=REAR=TEMP

ELSE

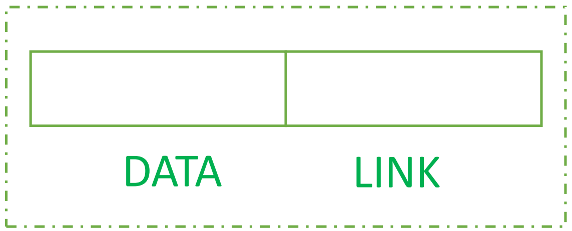
REAR→ LINK =TEMP

REAR=TEMP

[END OF IF]

STEP-4: EXIT

NODE STRUCTURE



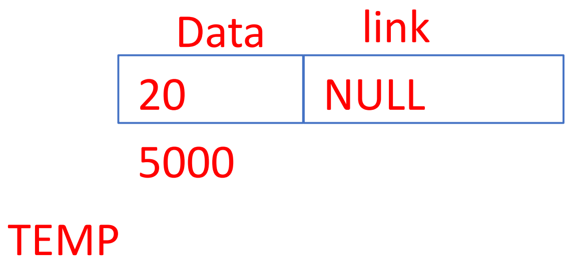
1. QINSERT_LINK(TEMP)

STEP-1: Allocate memory for the new node & name it as TEMP.

STEP-2: SET TEMP→ DATA = NUM

SET TEMP→ LINK = NULL

FRONT=REAR=NULL



STEP-3: IF FRONT = NULL

FRONT=REAR=TEMP

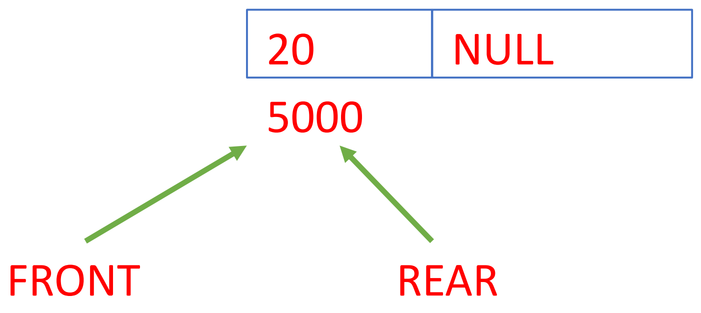
ELSE

REAR→ LINK =TEMP

REAR=TEMP

[END OF IF]

FRONT=REAR=5000



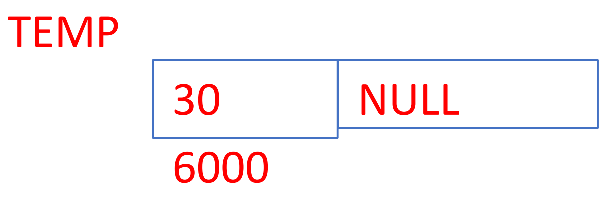
2. QINSERT_LINK(TEMP)

STEP-1: Allocate memory for the new node & name it as TEMP.

STEP-2: SET TEMP→ DATA = NUM

SET TEMP→ LINK = NULL

FRONT=REAR=TEMP



STEP-3: IF FRONT = NULL

FRONT=REAR=TEMP

ELSE

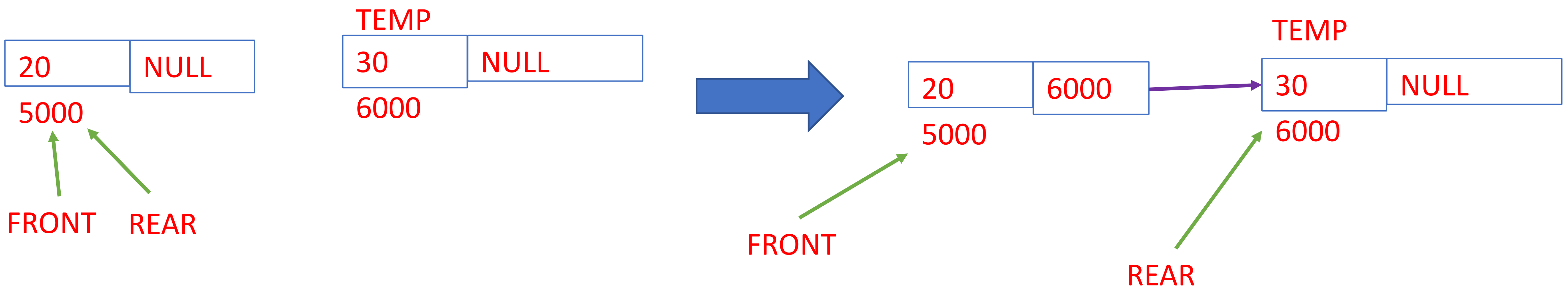
REAR→ LINK =TEMP

REAR=TEMP

[END OF IF]

FRONT=5000

REAR=6000



3. QINSERT_LINK(TEMP)

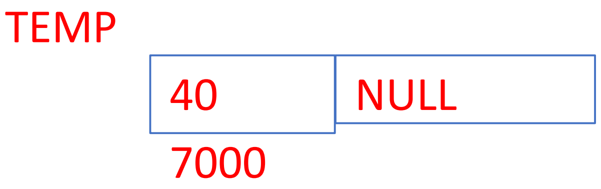
STEP-1: Allocate memory for the new node & name it as TEMP.

STEP-2: SET TEMP→ DATA = NUM

SET TEMP→ LINK = NULL

FRONT=5000

REAR=6000



STEP-3: IF FRONT = NULL

FRONT=REAR=TEMP

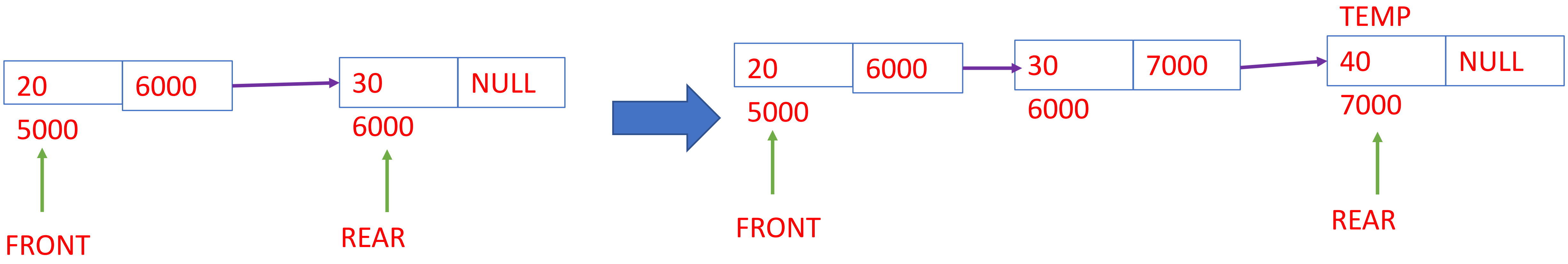
ELSE

REAR→ LINK =TEMP

REAR=TEMP

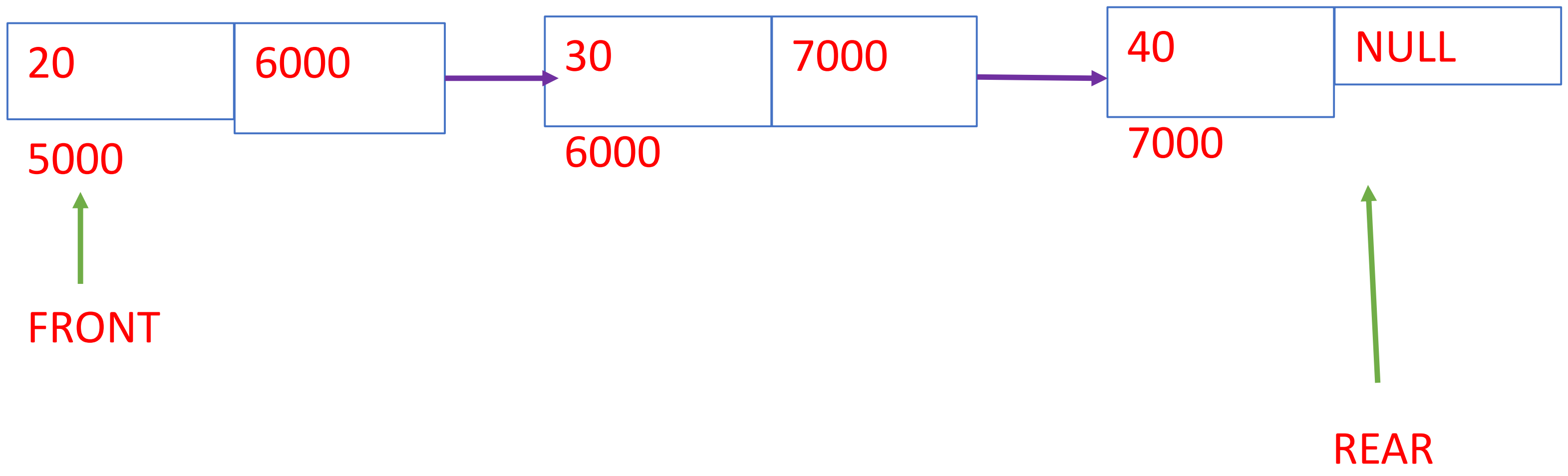
[END OF IF]

FRONT=5000
REAR=7000



FRONT=5000

REAR=7000



Delete an element in queue using linked list

STEP-1: If FRONT = NULL

write "underflow"

go to step 3.

[End of if]

STEP-2: SET TEMP = FRONT

FRONT=FRONT→ LINK

IF FRONT = NULL

REAR=NULL

STEP-3: EXIT

Delete an element in queue using linked list

1. QDELETE_LINK()

STEP-1: If FRONT = NULL

write "underflow"

go to step 3.

[End of if]

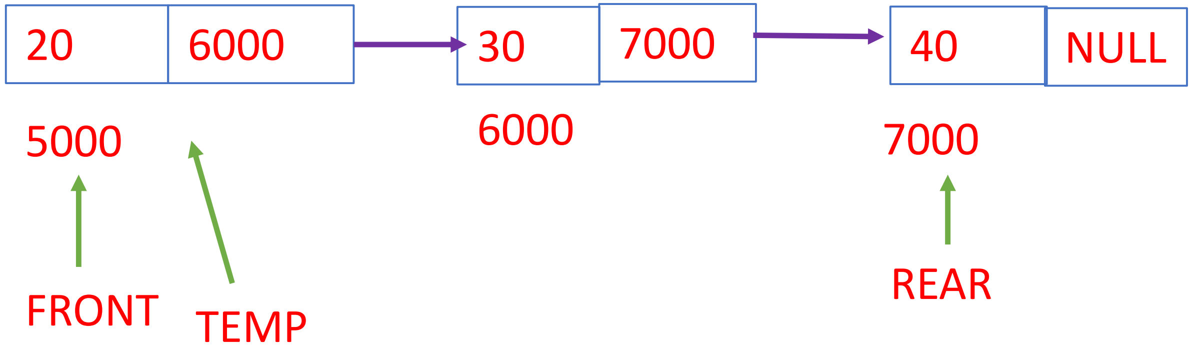
STEP-2: SET TEMP = FRONT

FRONT=FRONT→ LINK

IF FRONT = NULL

REAR=NULL

STEP-3: EXIT



Delete an element in queue using linked list

STEP-1: If FRONT = NULL

write "underflow"

go to step 4.

[End of if]

STEP-2: SET TEMP = FRONT

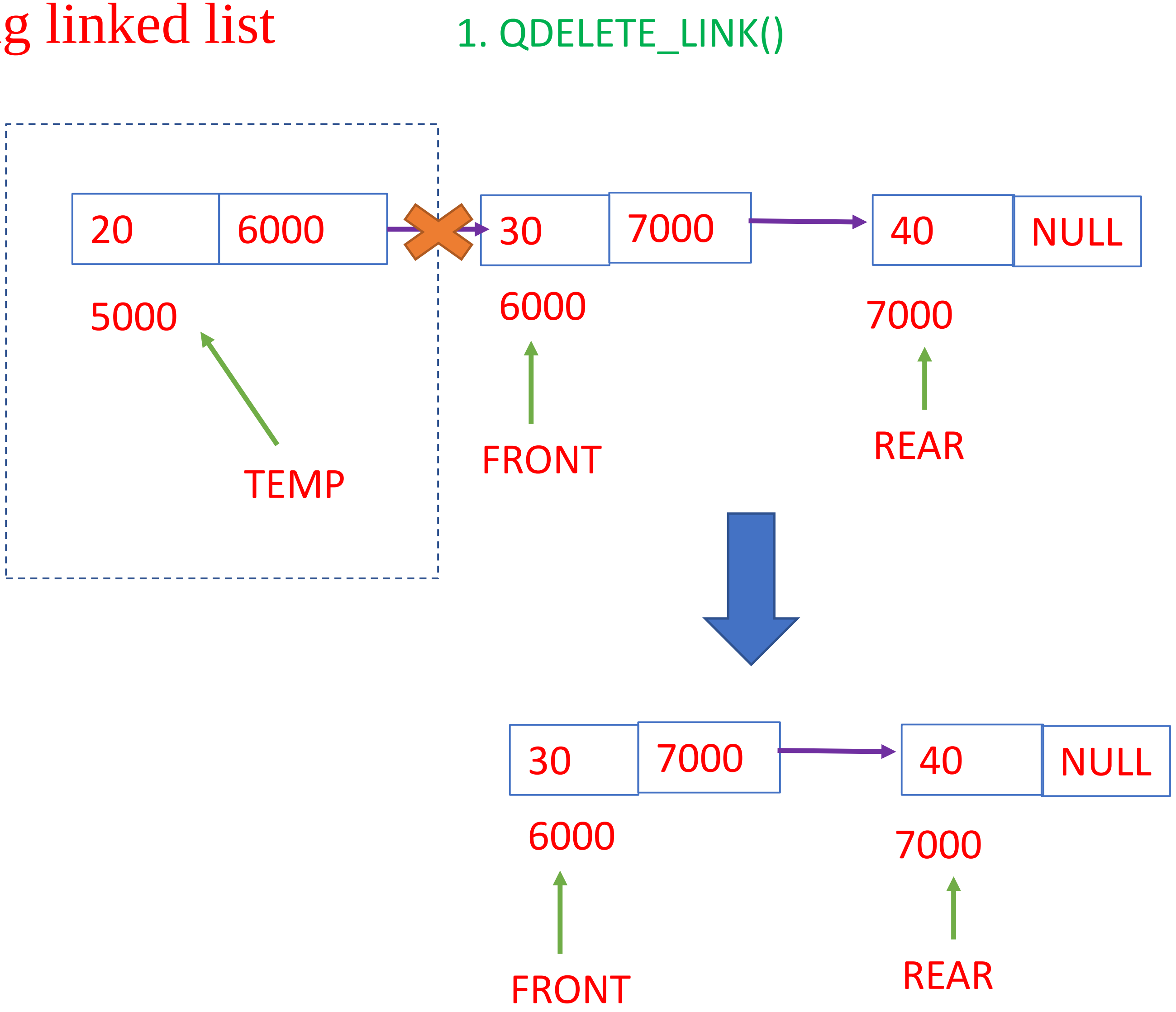
FRONT=FRONT→ LINK

IF FRONT = NULL

REAR=NULL

STEP-3: FREE(TEMP)

STEP-4: EXIT



2. QDELETE_LINK()

Delete an element in queue using linked list

STEP-1: If FRONT = NULL

write “underflow”

go to step 4.

[End of if]

STEP-2: SET TEMP = FRONT

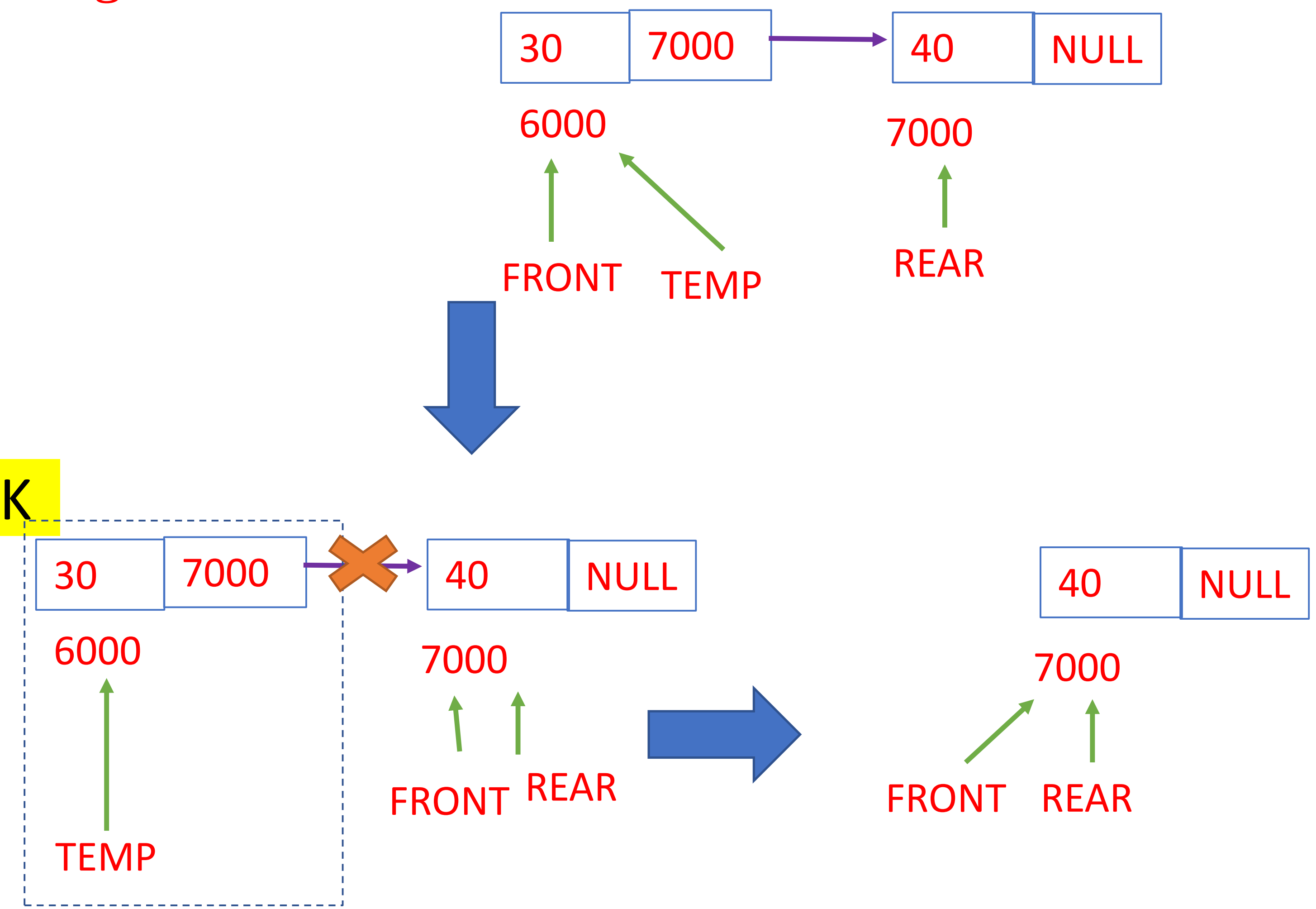
FRONT=FRONT→ LINK

IF FRONT = NULL

REAR=NULL

STEP-3: FREE(TEMP)

STEP-4: EXIT



3. QDELETE_LINK()
4. QDELETE_LINK()

Delete an element in queue using linked list

STEP-1: If FRONT = NULL
write "underflow"
go to step 4.
[End of if]

STEP-2: SET TEMP = FRONT
FRONT=FRONT→ LINK
IF FRONT = NULL
REAR=NULL

STEP-3: FREE(TEMP)

STEP-4: EXIT